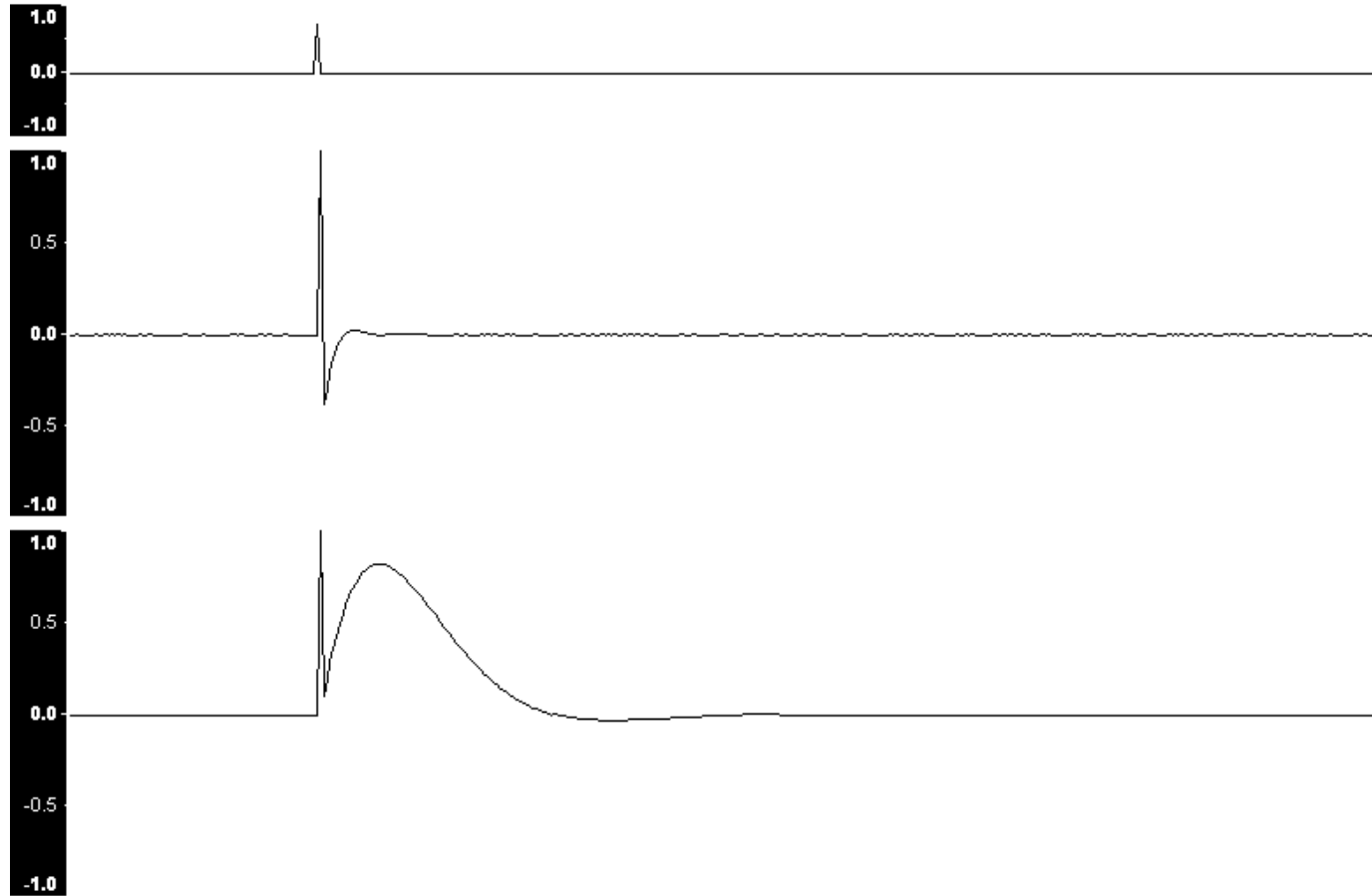


Einschwingungsverhalten

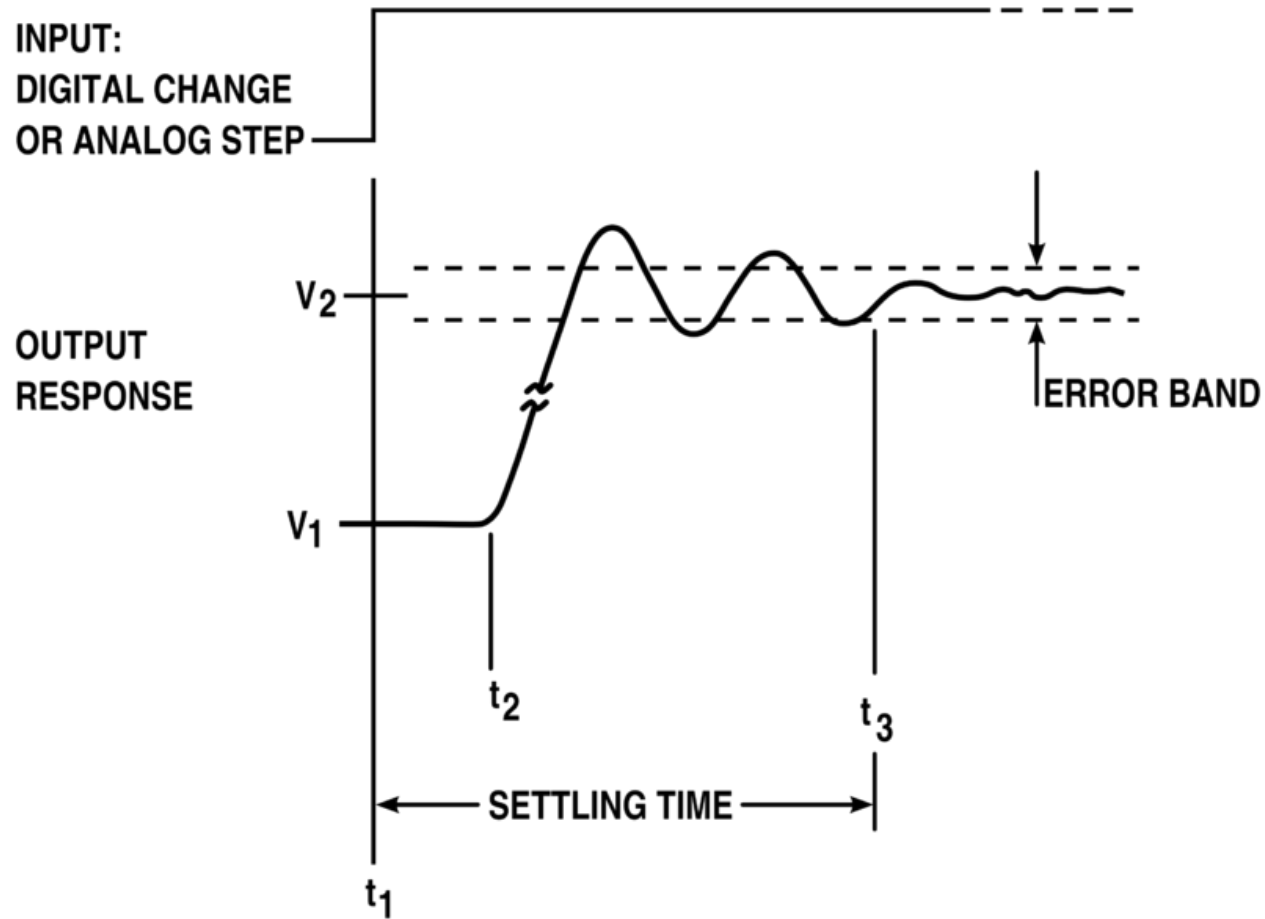
Präsentiert von:

Tim Kierot, Jens Ramermann, Andreas Hiebert

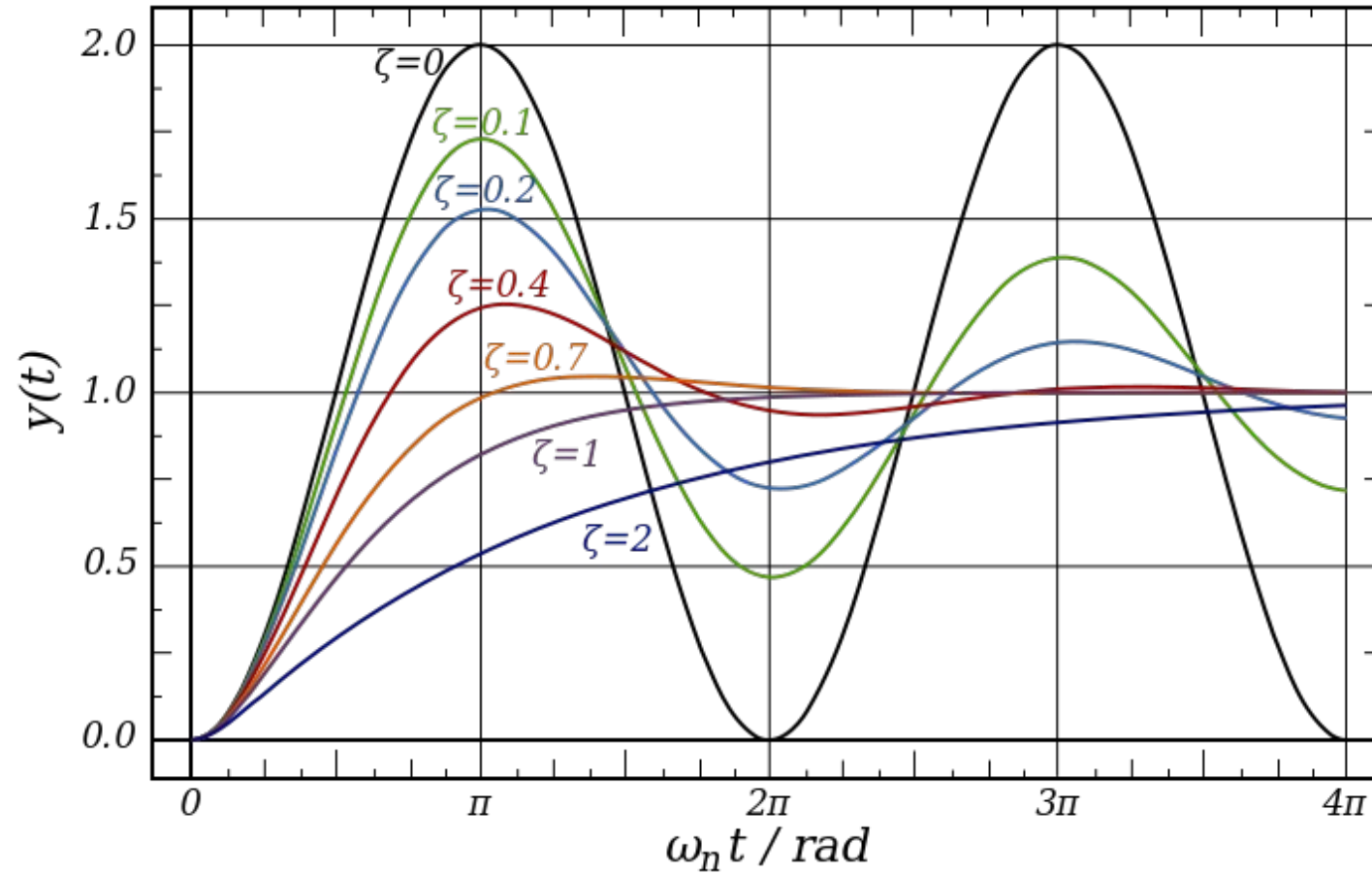
Impulsantwort



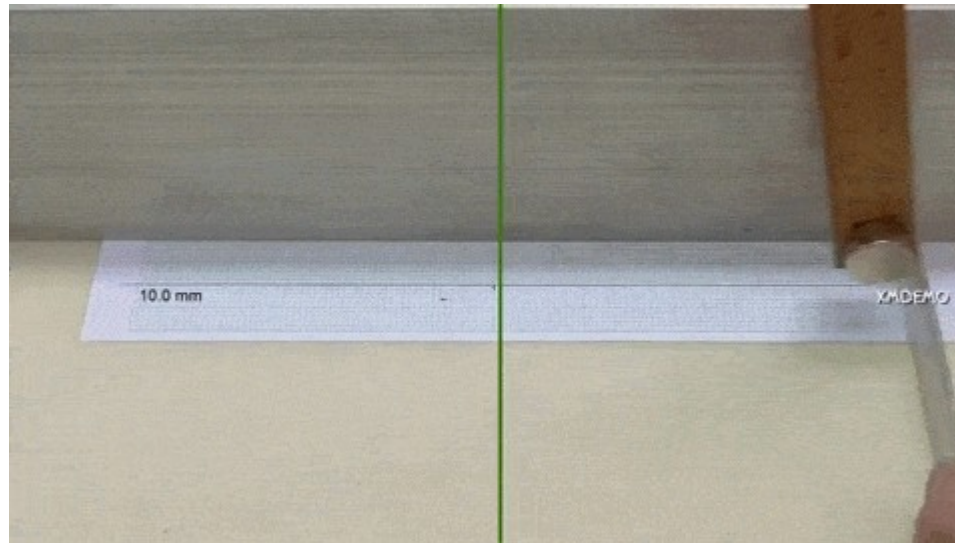
Schrittantwort



Dämpfungsmaß

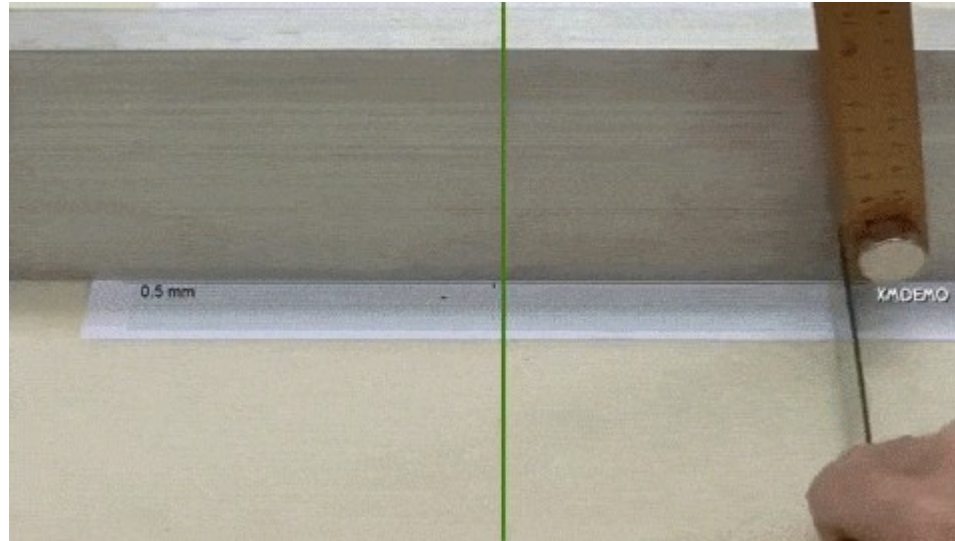


Gedämpftes System (Schwach) $0 < \zeta < 1$



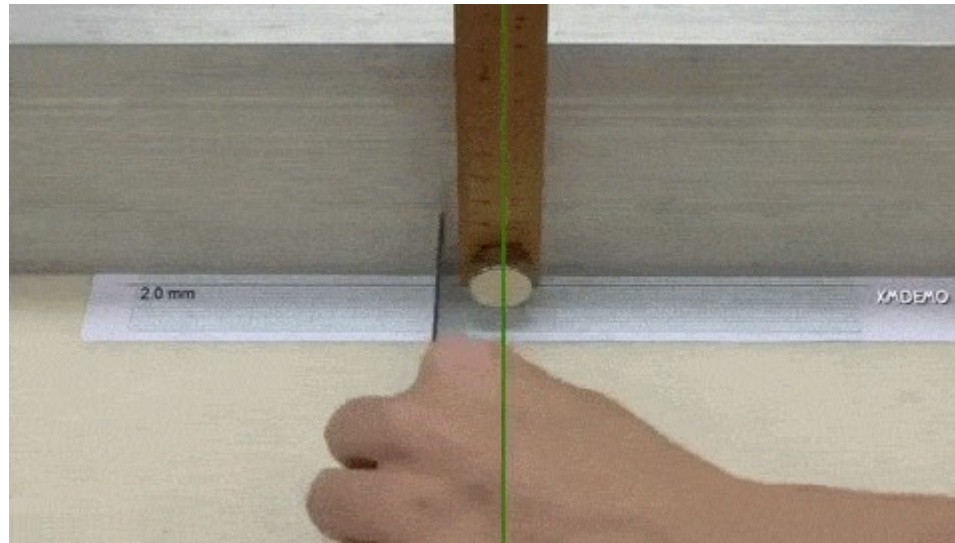
Schwingt bis Amplitude Null erreicht.

Aperiodische Lösung (Nicht schwingend) $\zeta > 1$



Kehrt direkt zu Null zurück ohne Überspringen.

Aperiodischer Grenzfall (Kritisch) $\zeta = 1$



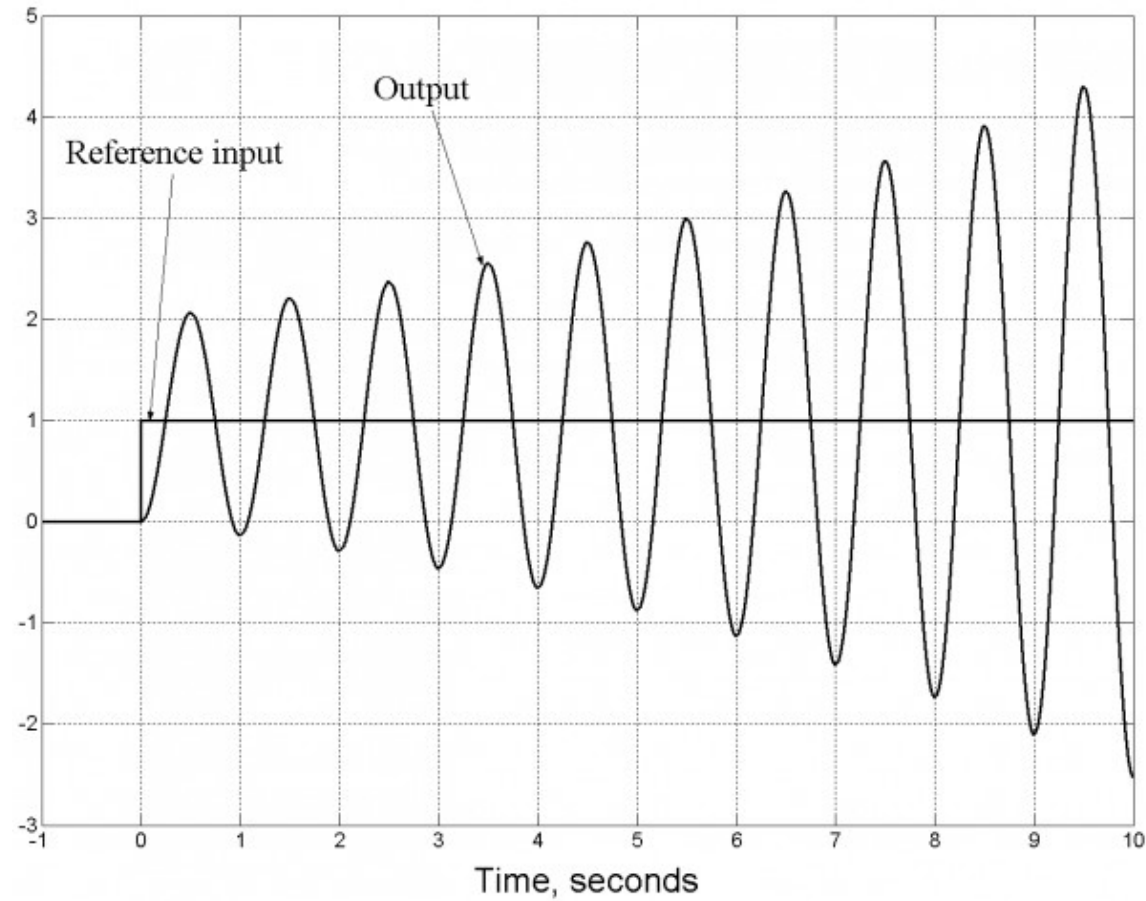
Kehrt direkt zu Null zurück so schnell es geht.

Ungedämpftes System (Grenzstabil) $\zeta = 0$



Dauerhaftes konstantes Schwingen mit natürlicher Resonanzfrequenz.

Instabiles System



Wächst über alle Grenzen hinaus.

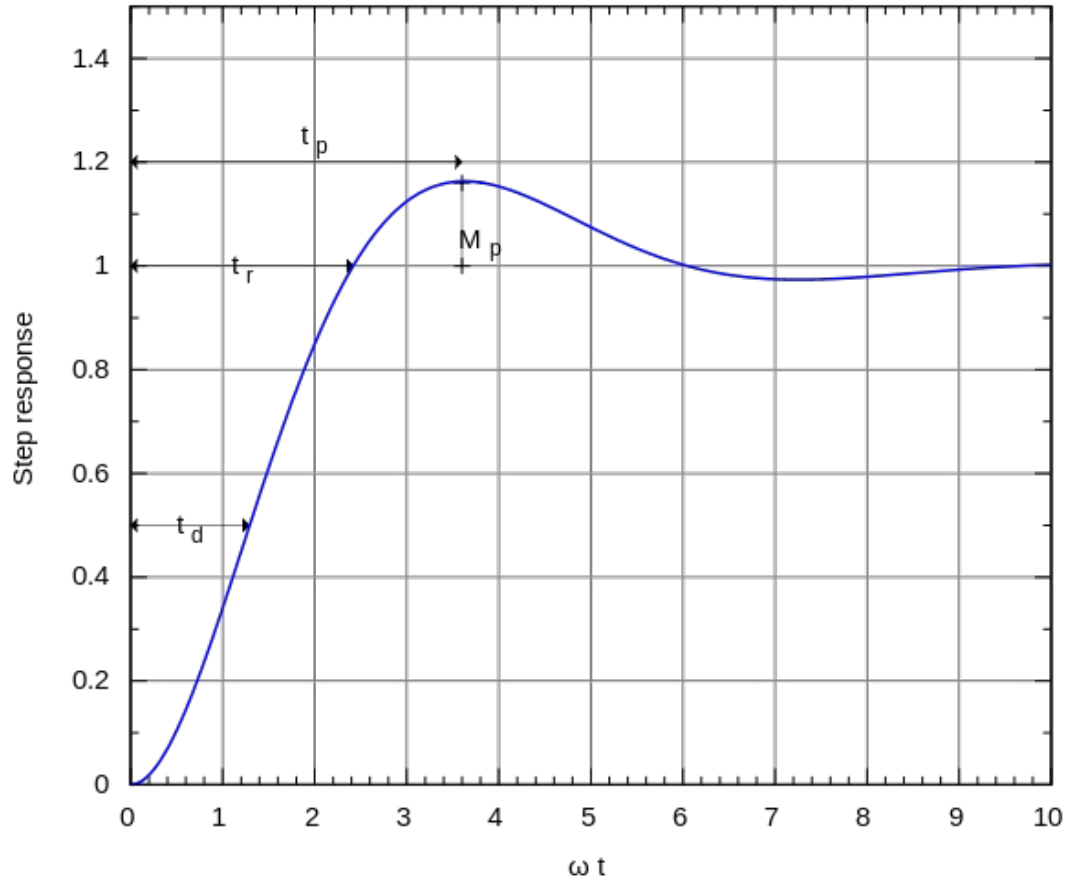
Eigenschaften

Rise time $t(r)$

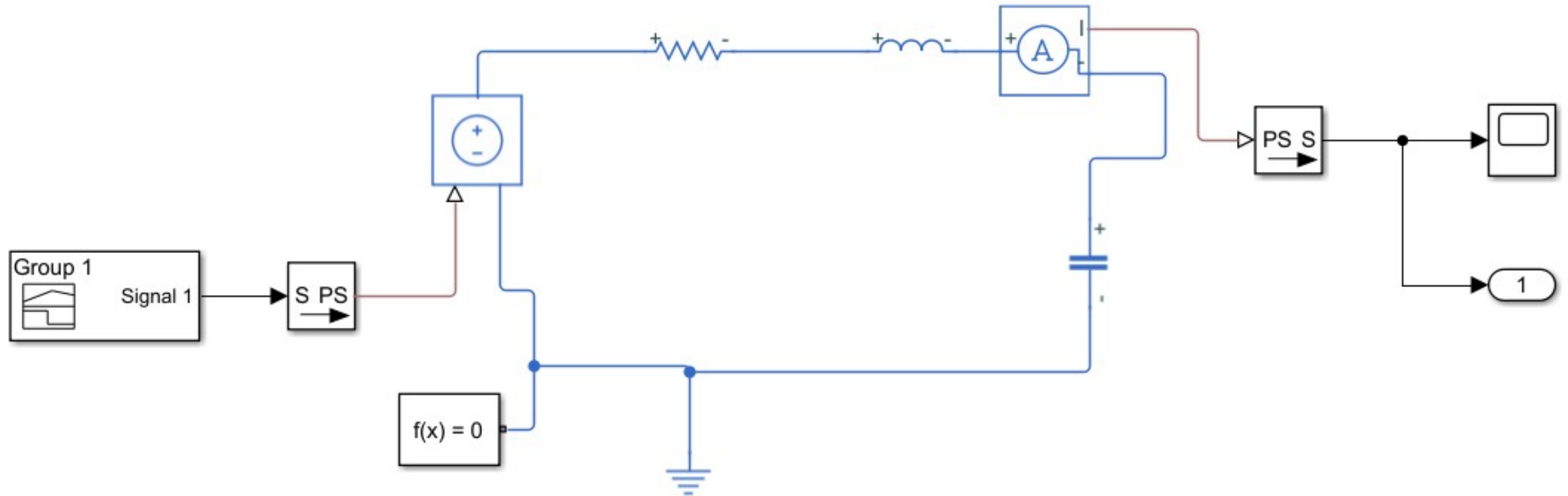
Peak time $t(p)$

Delay time $t(d)$

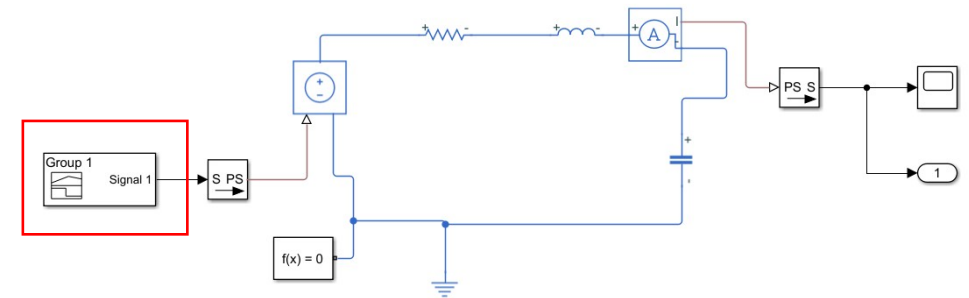
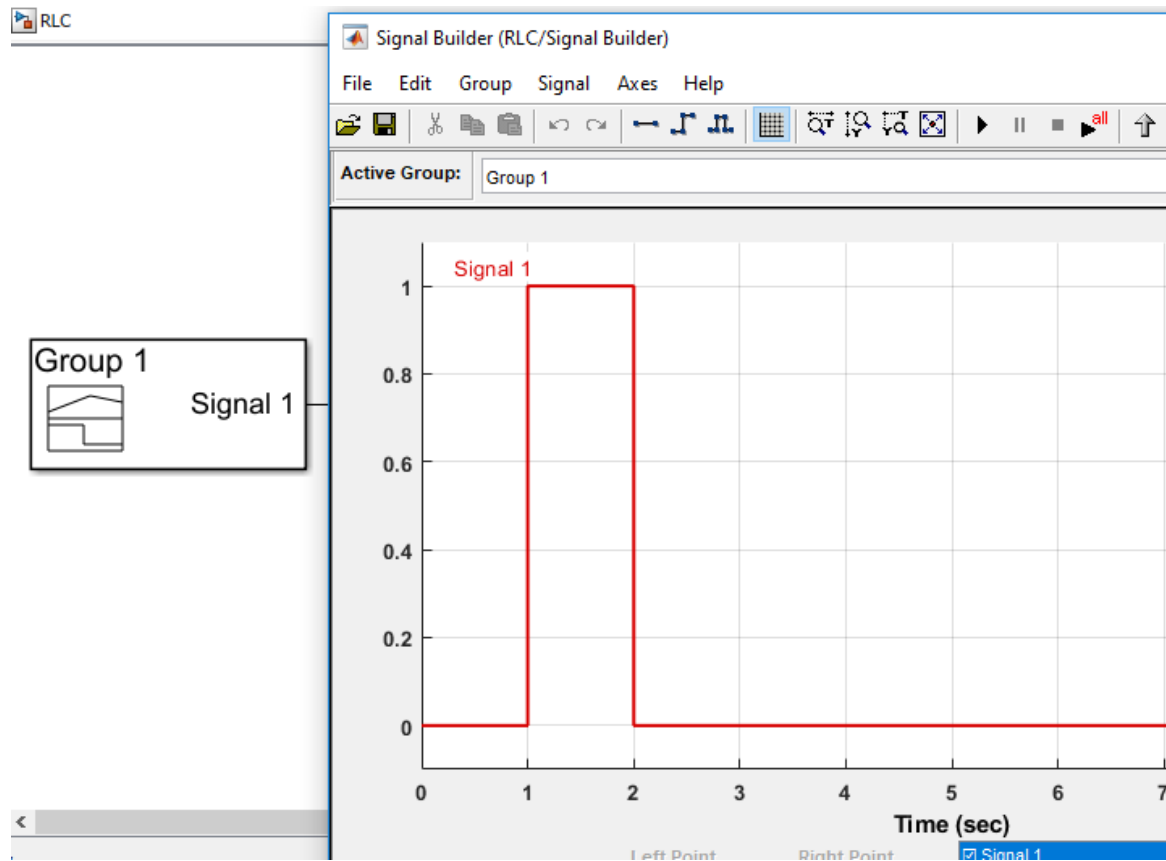
Maximum Overshoot $M(p)$



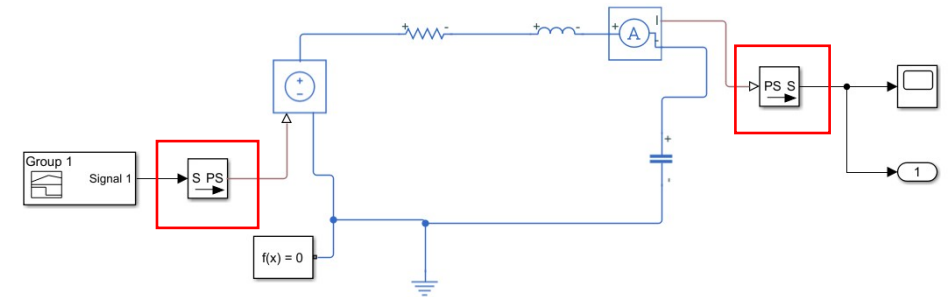
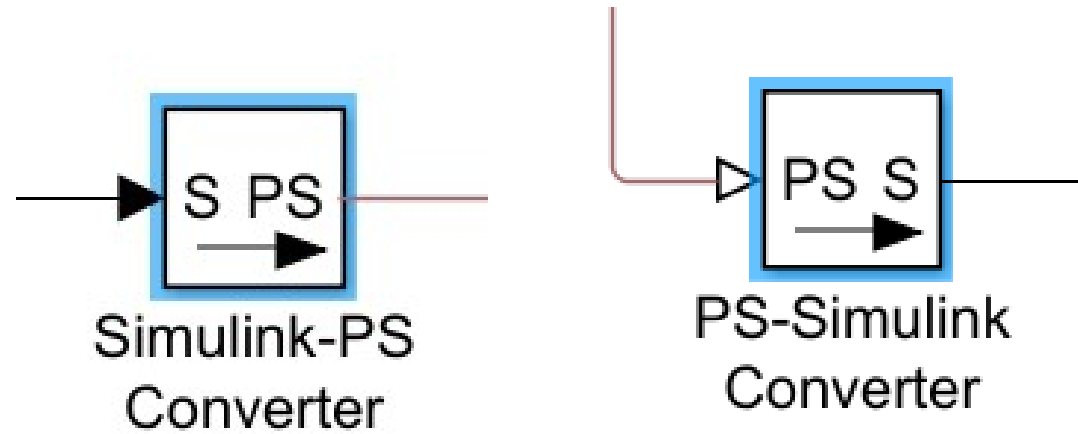
Unser RLC Modell



Signal Builder

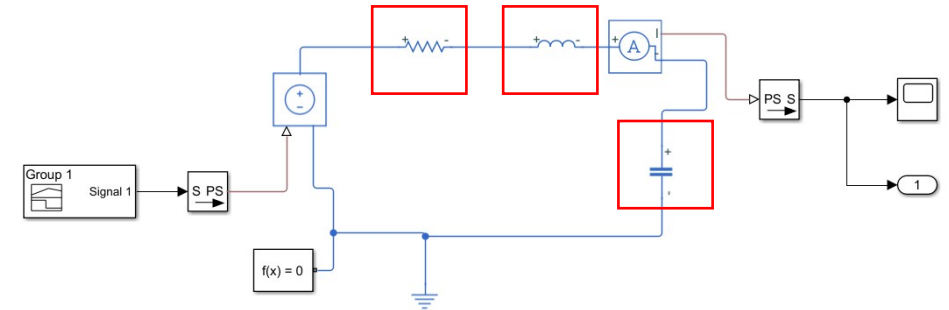


PS-Simulink / Simulink-PS Converter

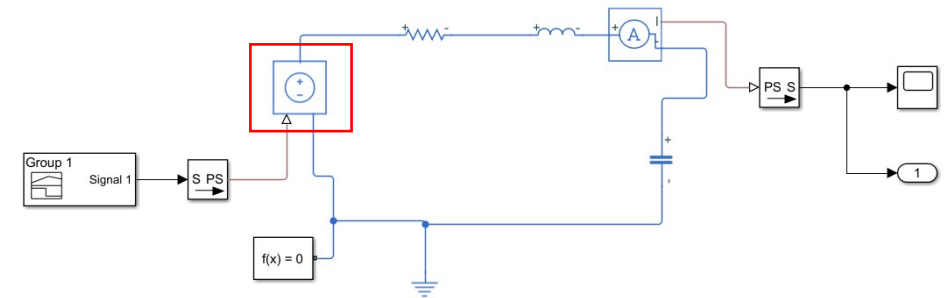
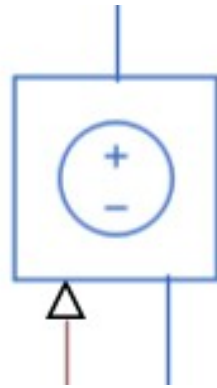


Elektrische Blöcke

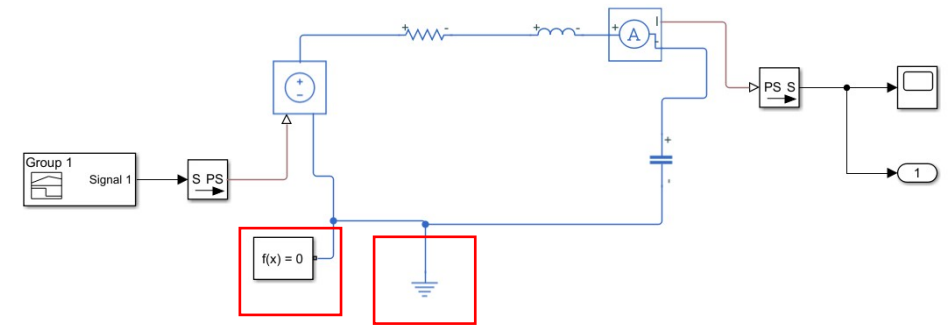
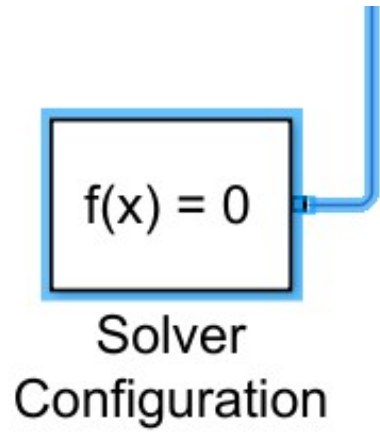
- Widerstand: (hier 1 Ohm)
- Spule: (hier 1 Henry)
- Kondensator: (hier 0,2 Farad)



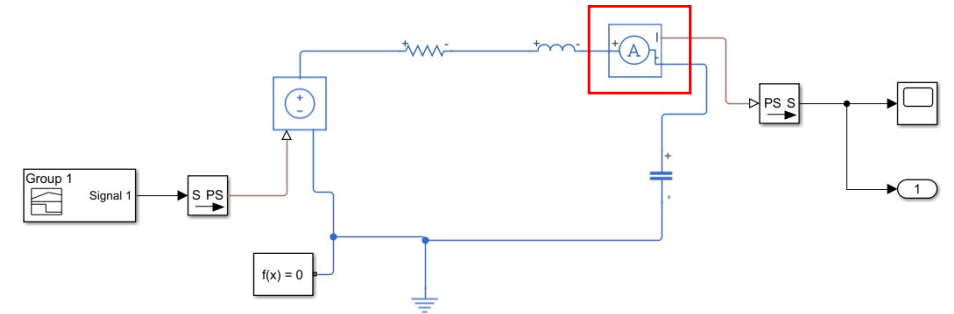
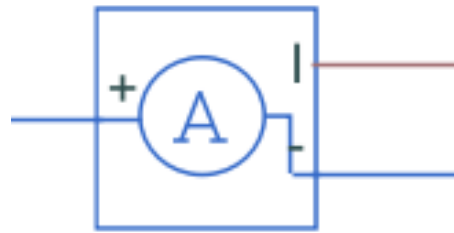
Controlled Voltage Source



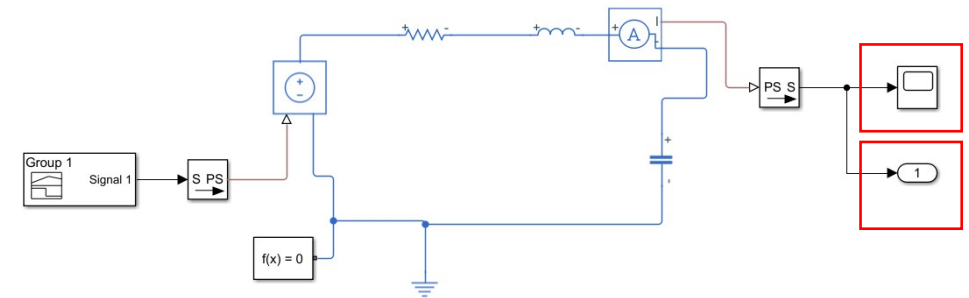
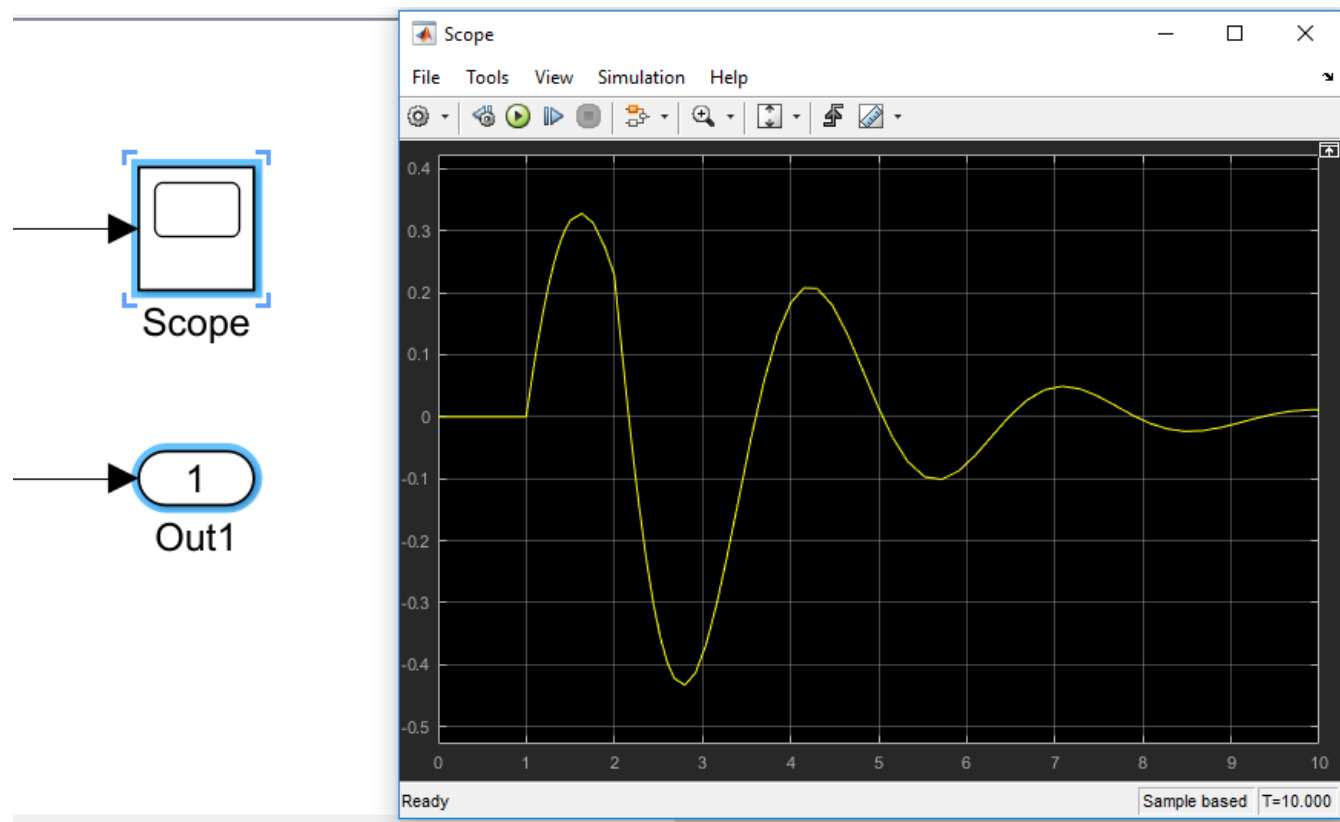
Kritische Blöcke

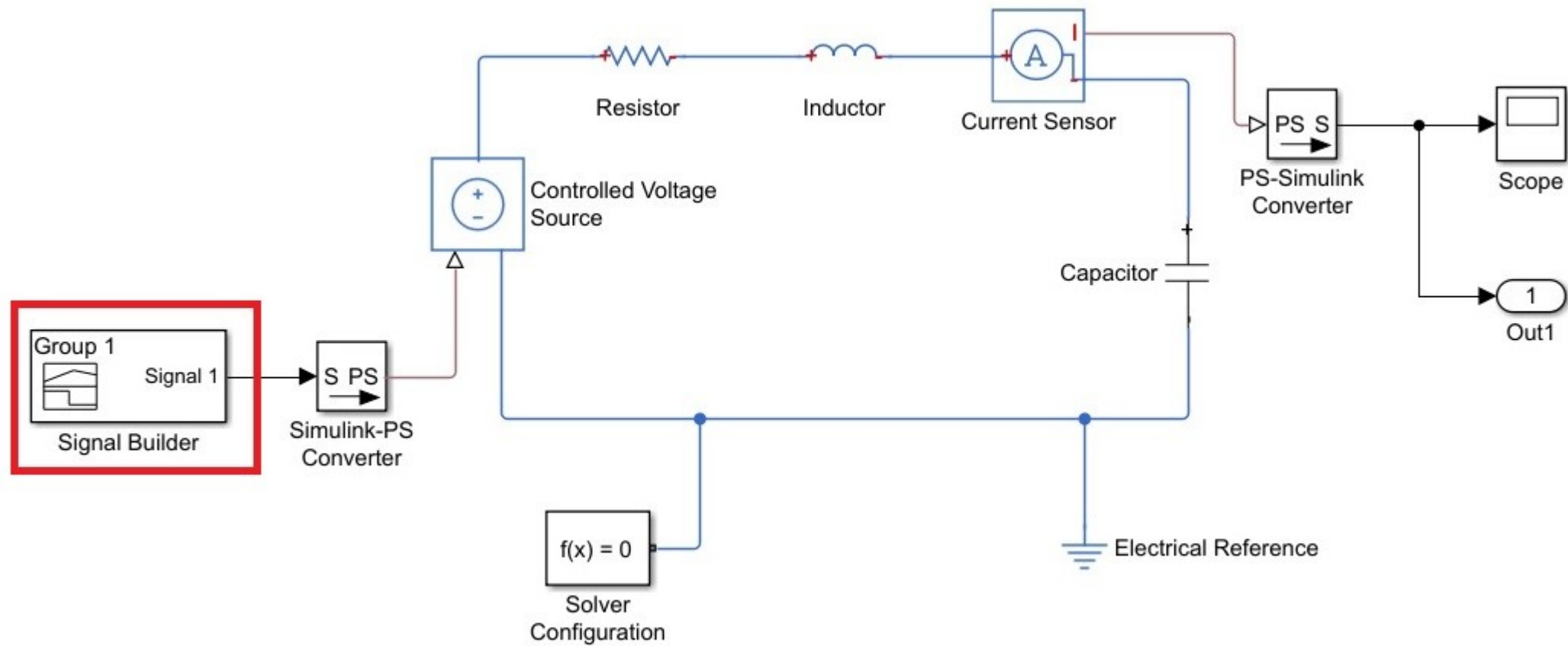


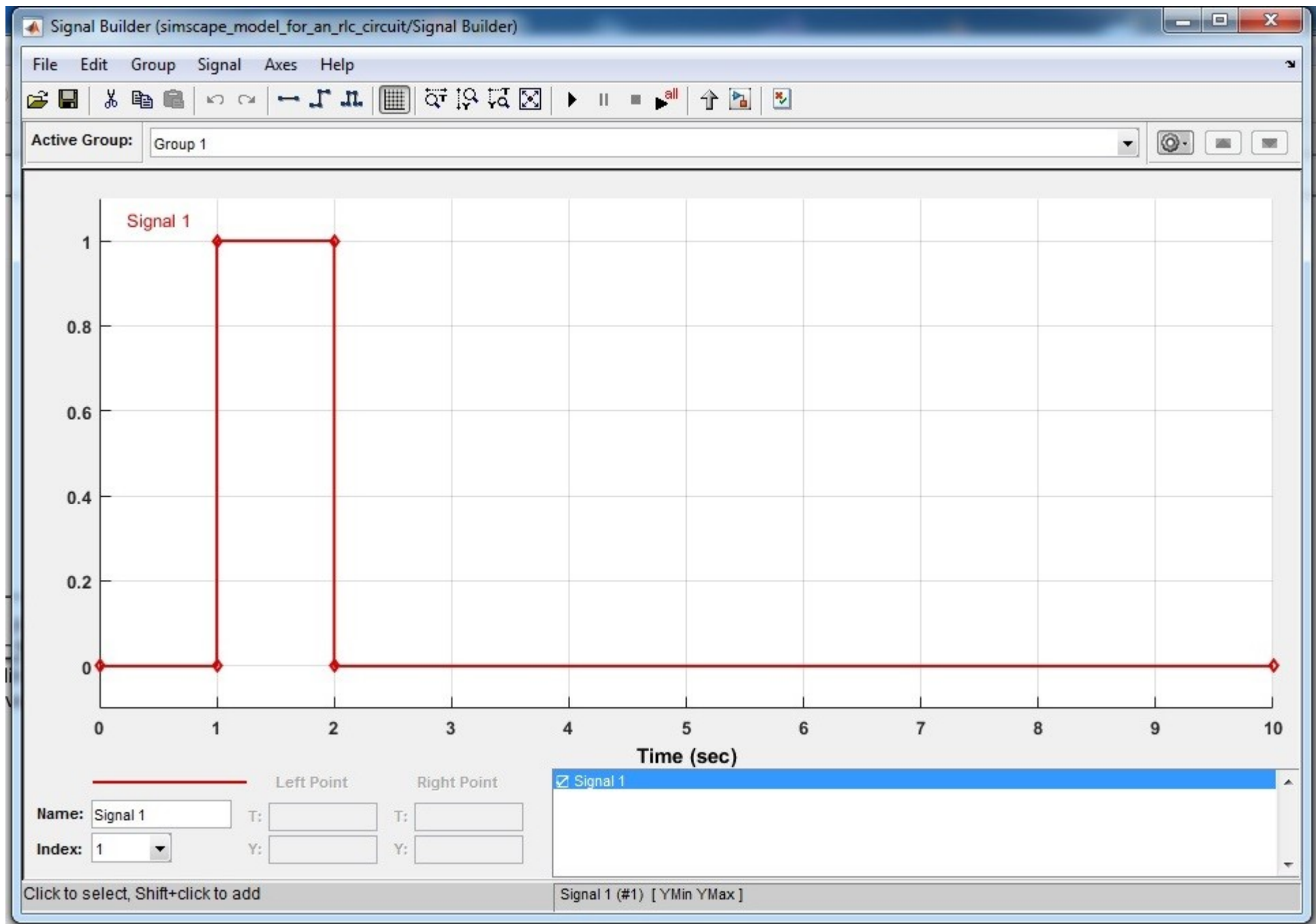
Current Sensor

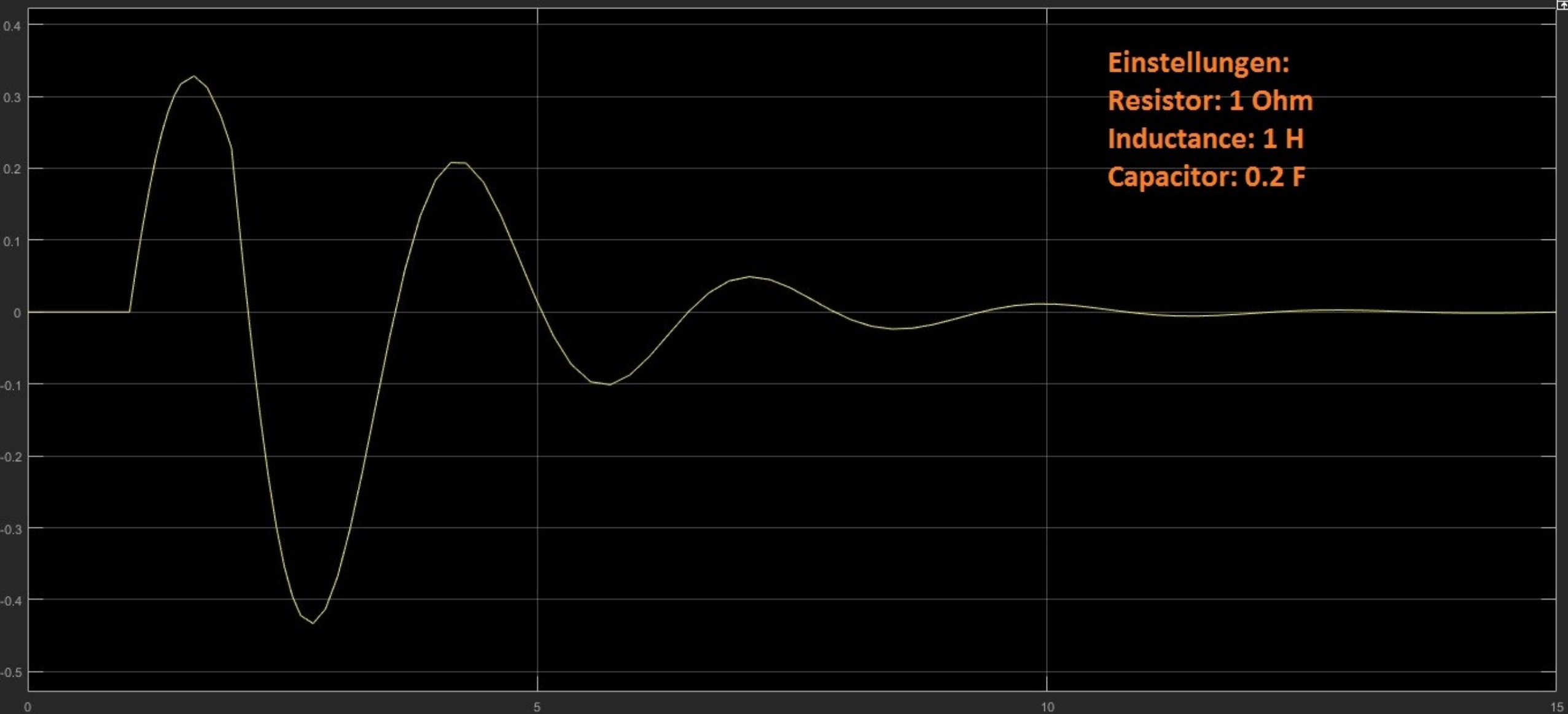


Ausgangsblöcke

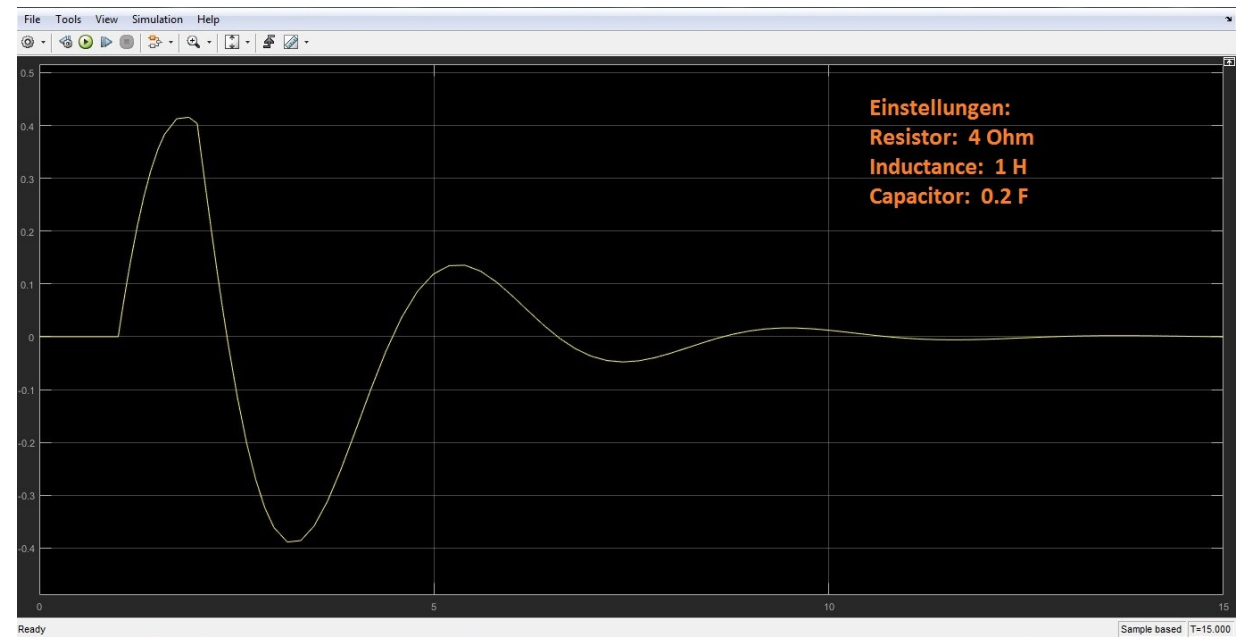
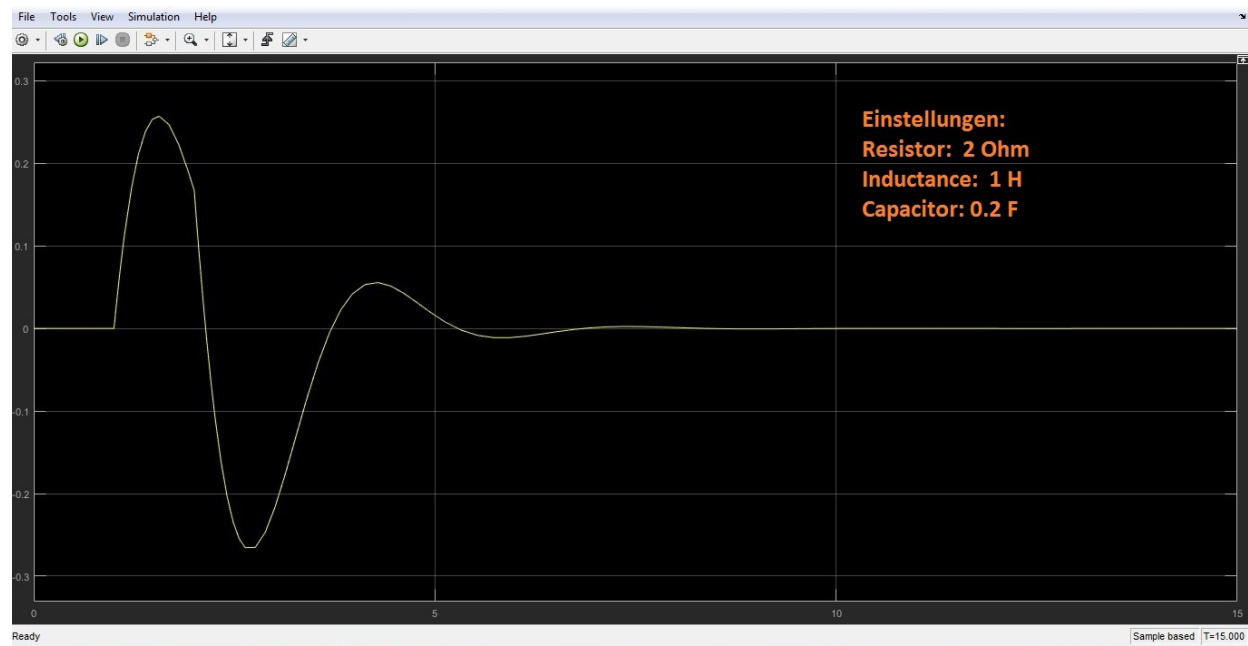
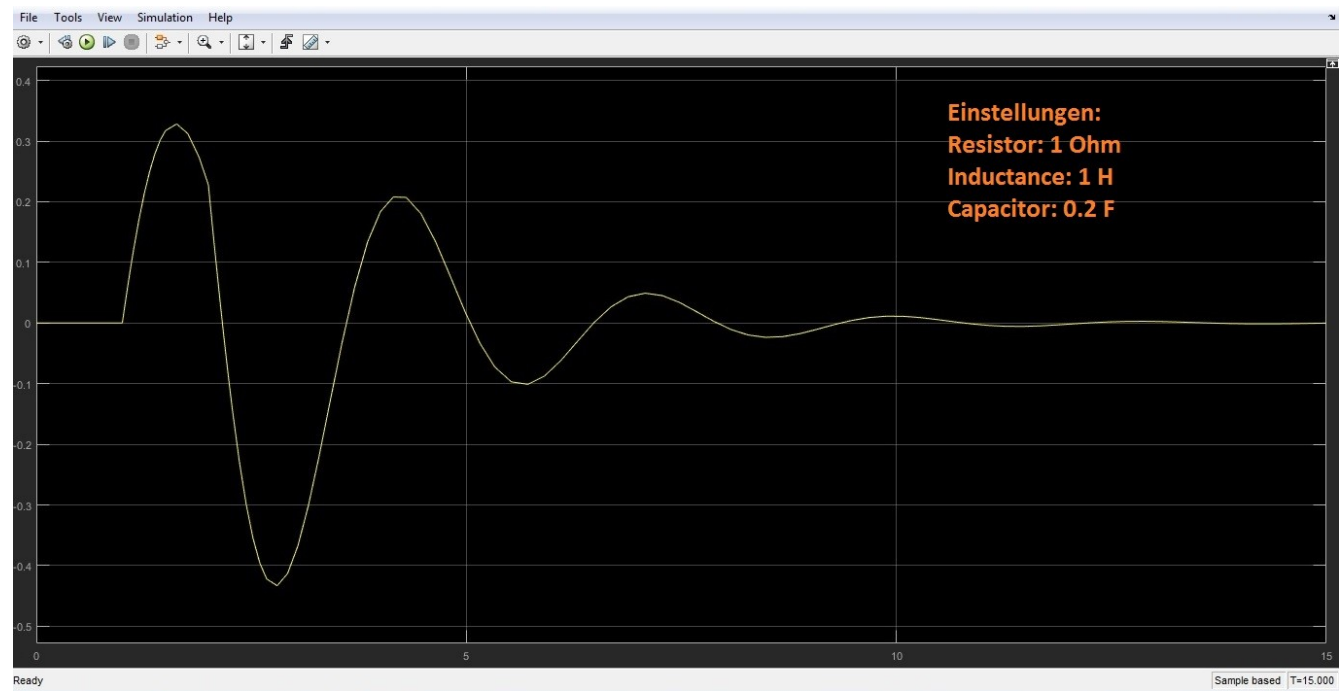


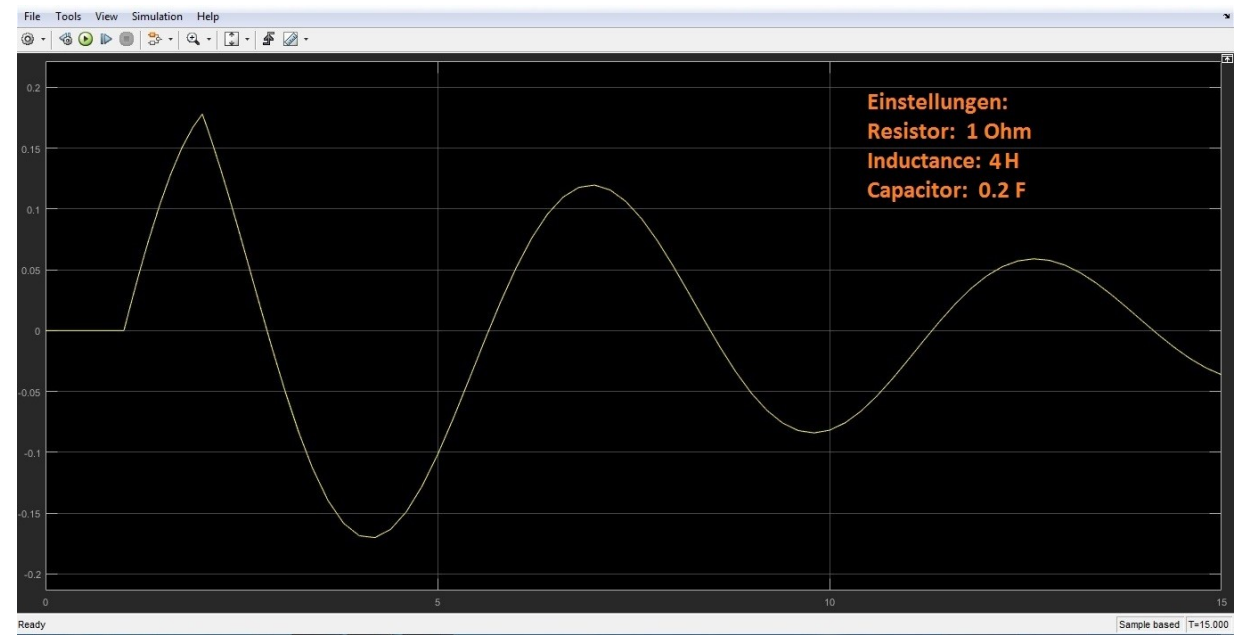
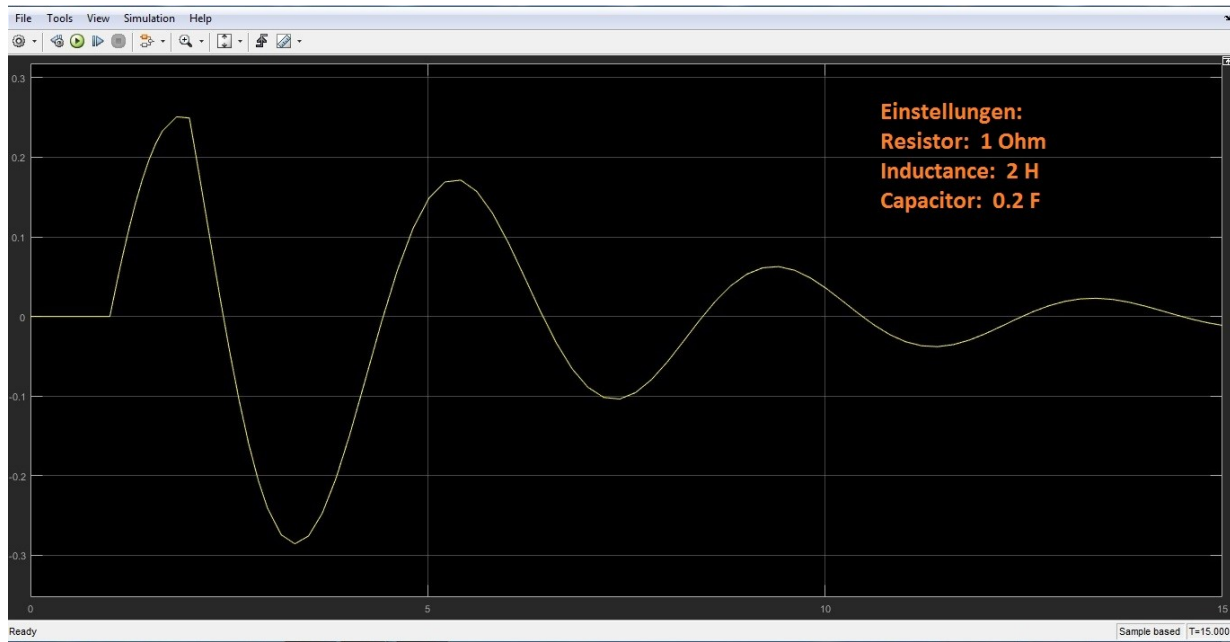
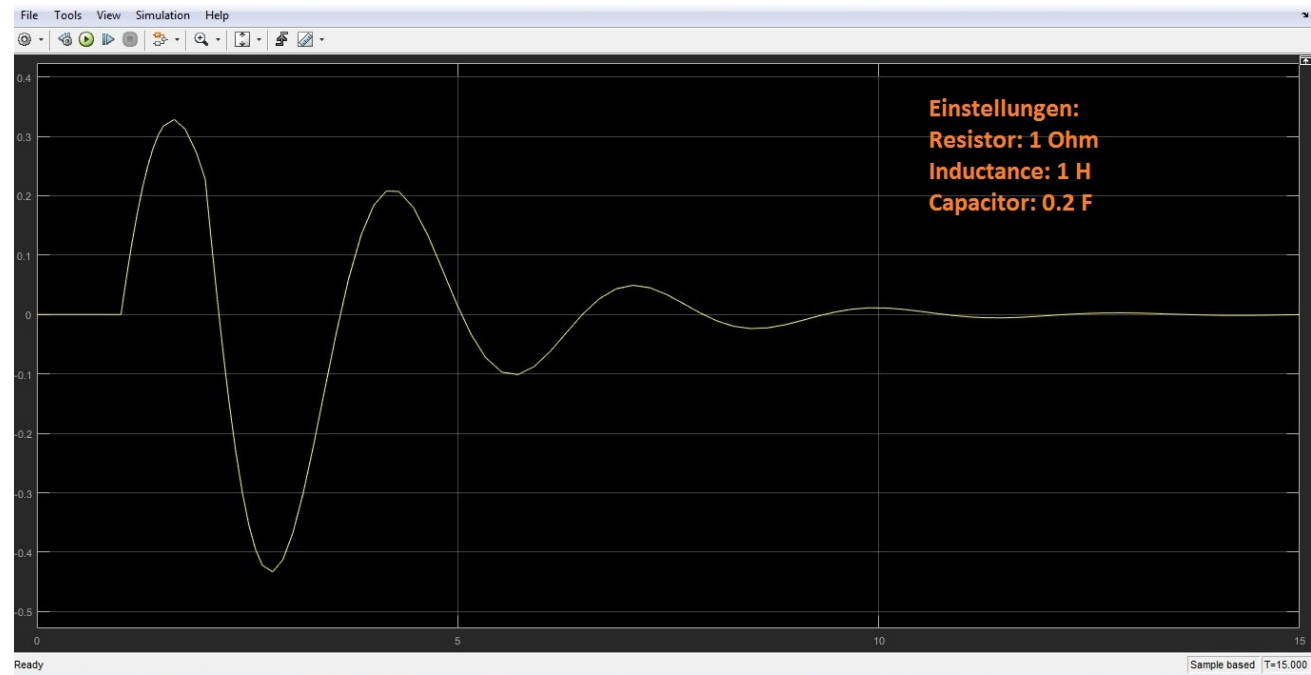


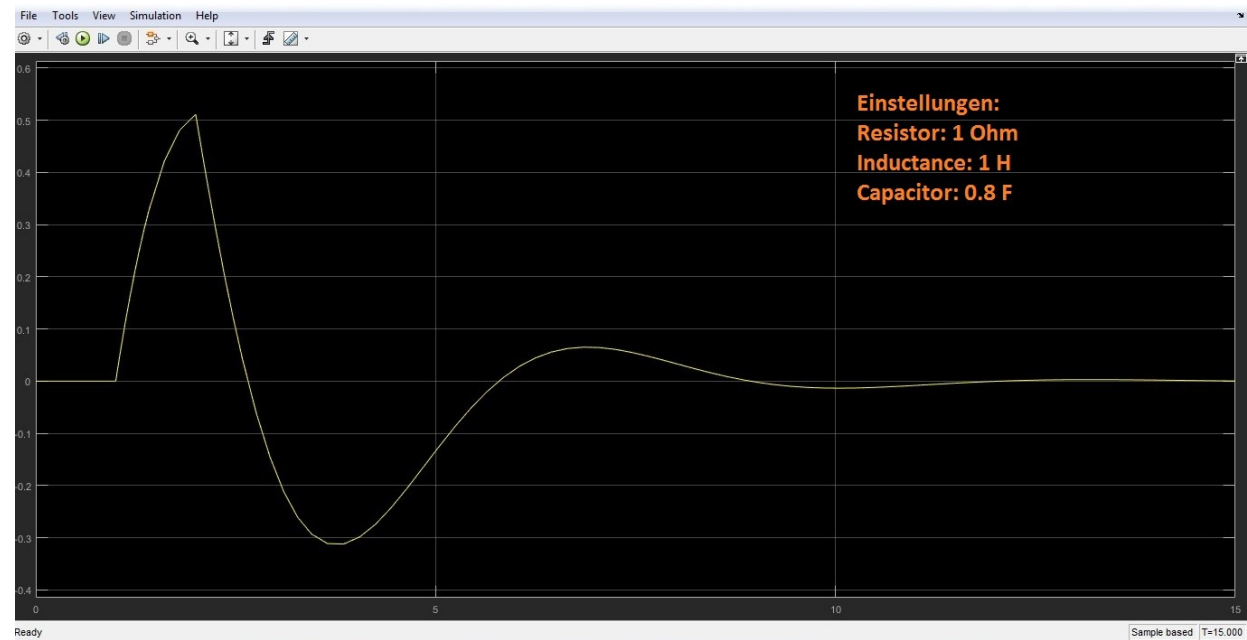
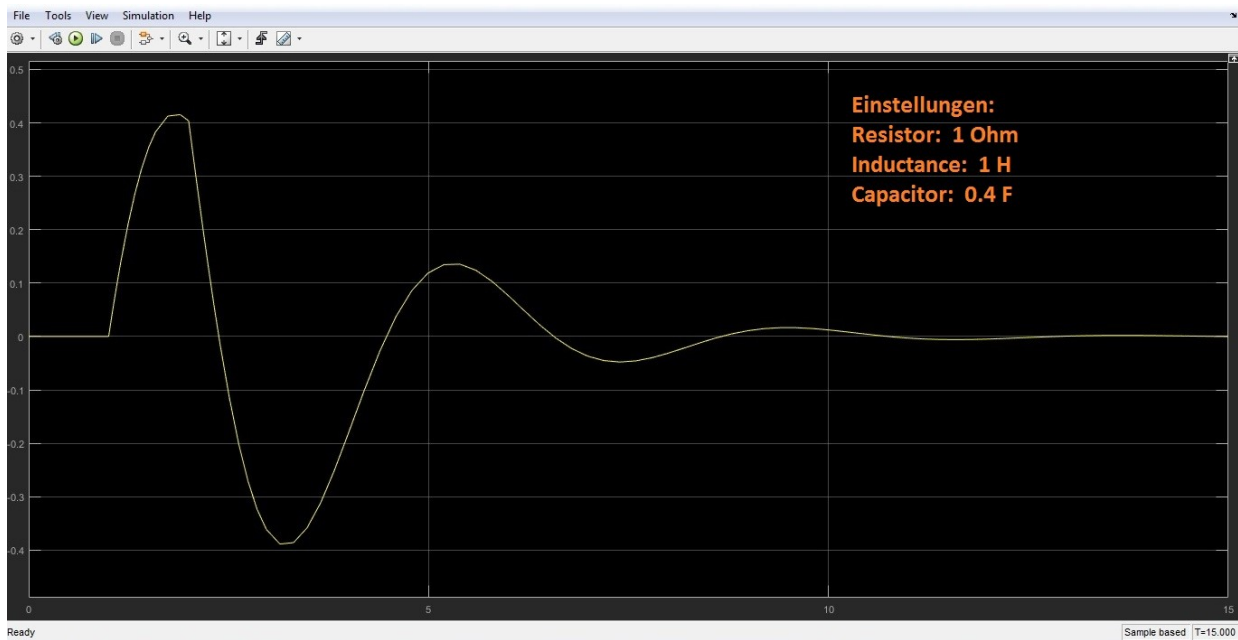
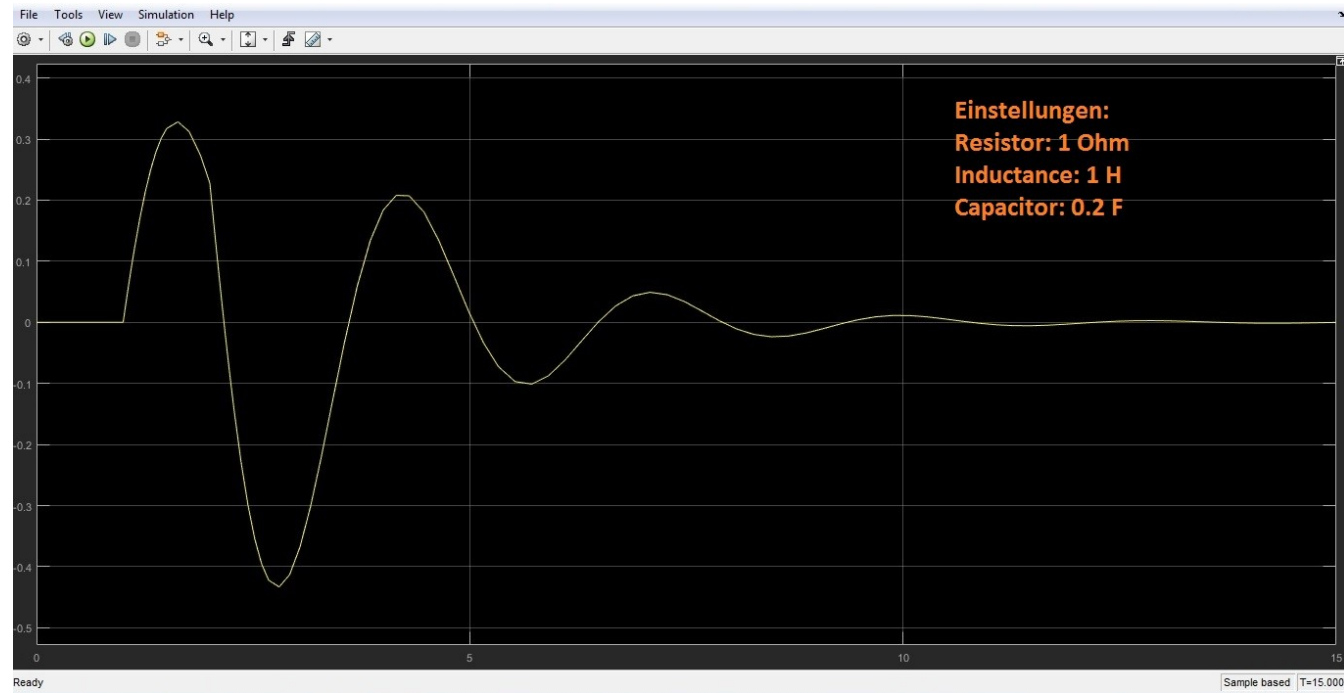




Einstellungen:
Resistor: 1 Ohm
Inductance: 1 H
Capacitor: 0.2 F







Vielen Dank für Eure Aufmerksamkeit

Gibt es Fragen?

Verständnisfragen zu unserer Präsentation

- 1) Wie hoch ist das Dämpfungsmaß eines Systems mit konstanten Schwingungen?

- 2) Zeichnen Sie ein: (Graph auf Board zeichnen)
 - a) Rise time
 - b) Peak time
 - c) Delay time
 - d) Maximum Overshoot

Fragen

3) Welche Blöcke sind kritisch für das RLC Model?

4) Welche Blöcke verbinden die Simscape und Simulink Systeme um die Kommunikation zwischen den beiden System zu ermöglichen?

Fragen

5) Welche Funktion hat der Block "Signal Builder"?

(Was kann man da einstellen?)

6) Was passiert, wenn man den Wert beim Inductance verdoppelt, vervierfacht?

Quellen

- Impulsantwort: https://en.wikipedia.org/wiki/Impulse_response
- Schrittantwort: https://en.wikipedia.org/wiki/Step_response
- Dämpfungsgrad: https://en.wikipedia.org/wiki/Damping_ratio
- Dämpfungstypen: <https://imgur.com/gallery/NErmd>
- Instabiles System: <http://www.ledin.com/control-systems-basics/>